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**Thirteenth Section. General Theory of the Equation of the
Fifth Degree**

§106. Statement of the Problem

In §44 we saw that, beyond the question of the Galois group of an algebraic equation, there is a still more far-reaching problem: to ask for a linear substitution group of as small a number of dimensions as possible, to whose problem of forms the given equation can be reduced. In metacyclic equations, and in particular in the general equations of the third and fourth degrees, this number of dimensions is equal to one. This is precisely another way of saying that these equations are solvable by radicals. The next equations to be investigated are therefore the equations of the fifth degree; and it will appear that the solution of the general equation of the fifth degree leads to a binary linear substitution group, namely to the icosahedral problem.

Connected with this is another question. The general equation of the fifth degree contains five coefficients, which may be regarded as independent variables; hence a root of such an equation is an algebraic function of five variables. But the number of these variables may already be reduced by simple linear substitutions. By a Tschirnhausen transformation, for example to Jerrard's form, the equation can even be made to depend on only one variable coefficient, that is, on one parameter. Thus, by mediation of equations which do not reach the fifth degree, one may make the root of a general equation of the fifth degree depend on an algebraic function of one variable.

The question to which we are led is therefore this: how can the solution of an algebraic equation, whose coefficients depend on a certain number of variables, be reduced to an equation with as small a number of parameters as possible; and, in particular, under what circumstances and by what auxiliary means can it be reduced to equations with only one parameter?

We accordingly pose the following question: under what hypotheses do resolvents with only one parameter exist for a general equation of degree n ?

Let

$$x_0, x_1, \dots, x_{n-1}$$

be a system of n independent variables, and let

$$u = \psi(x_0, x_1, \dots, x_{n-1}) \tag{1}$$

be a rational function of these variables. Under the substitutions of the alternating

group on the n letters x , suppose that u is changed into the mutually distinct functions

$$u, u_1, u_2, \dots, u_{\nu-1}, \quad (2)$$

where ν may be equal to or smaller than the degree of the alternating group. Let z denote a function of the variables x which remains unchanged under the alternating group. We ask when there exists a rational equation of degree ν in u ,

$$F(u, z) = 0, \quad (3)$$

which is identically satisfied by the ν functions just written. The coefficients in this equation must be independent of the x and hence must be pure numbers. For the time being we impose no restriction on the numerical field of rationality.

If $\nu > 1$, the equation $F(u, z) = 0$ is a resolvent of the equation of degree n whose roots are the x 's, and it depends on the single parameter z . We exclude the case $\nu = 1$.

§107. Luroth's Theorem

We first need the following auxiliary theorem.

1. Let

$$u, u_1, u_2, \dots, u_{\nu-1}$$

be a system of rational functions of one variable t . Then there is a rational function ϑ of the $u, u_1, \dots, u_{\nu-1}$ such that $u, u_1, \dots, u_{\nu-1}$ can be expressed rationally by ϑ alone.

For the proof, write

$$u = \frac{\varphi(t)}{\psi(t)}, \quad u_1 = \frac{\varphi_1(t)}{\psi_1(t)}, \quad \dots, \quad u_{\nu-1} = \frac{\varphi_{\nu-1}(t)}{\psi_{\nu-1}(t)}, \quad (4)$$

where $\varphi(t), \psi(t)$ are integral functions of t without a common divisor, and similarly for the pairs $\varphi_i(t), \psi_i(t)$. We may also suppose that none of the functions $u, u_1, \dots, u_{\nu-1}$ is constant. Introduce, beside t , a second variable τ and put

$$\varphi(t)\psi(\tau) - \varphi(\tau)\psi(t) = \Phi(t, \tau) = -\Phi(\tau, t),$$

$$\varphi_i(t)\psi_i(\tau) - \varphi_i(\tau)\psi_i(t) = \Phi_i(t, \tau) = -\Phi_i(\tau, t).$$

As functions of t all these expressions vanish for $t = \tau$; consequently they have a greatest common divisor, of positive degree in t . Let this common divisor be

$$R(t, \tau). \quad (5)$$

This function is rational also in τ . It may be normalized so that τ occurs neither in a denominator nor in a superfluous factor. Then, upon interchanging t and τ , $R(t, \tau)$ can change at most by sign. For the expressions above are alternating in the two variables and are divisible by $R(t, \tau)$, and also by $R(\tau, t)$. Hence the two functions divide one another and differ only by a constant factor, which must be ± 1 .

It remains to observe that, for undetermined τ , none of the functions $\Phi_i(t, \tau)$, considered as functions of t , contains a repeated factor. Suppose, for example, that

$$\Phi(t, \tau) = \varphi(t)\psi(\tau) - \varphi(\tau)\psi(t),$$

$$\Phi'(t, \tau) = \varphi'(t)\psi(\tau) - \varphi(\tau)\psi'(t)$$

had a common divisor P as functions of t . Then P would divide

$$\psi(t)\Phi'(t, \tau) - \psi'(t)\Phi(t, \tau) = (\varphi'(t)\psi(t) - \varphi(t)\psi'(t))\varphi(\tau),$$

and therefore P could be assumed independent of τ . If now two values τ_1, τ_2 are chosen so that $\Phi(\tau_2, \tau_1)$ is non-zero, the identity obtained by eliminating the parameter shows that both $\varphi(t)$ and $\psi(t)$ would be divisible by P . This contradicts the relative primality of φ and ψ .

Thus $R(t, \tau) = -R(\tau, t)$ up to the chosen sign. To form $R(t, \tau)$ one may take the greatest common divisor of the functions

$$\varphi_h(t)\psi_h(\tau) - \varphi_h(\tau)\psi_h(t).$$

It follows that, apart from a factor independent of t , $R(t, \tau)$ is rationally expressible by $u, u_1, \dots, u_{\nu-1}$. If a, b are any fixed numerical values, put

$$\vartheta = \frac{R(b, t)}{R(a, t)}. \quad (6)$$

Then ϑ is a rational function of $u, u_1, \dots, u_{\nu-1}$.

For a new variable ξ the equation

$$R(a, \xi) - \vartheta R(b, \xi) = 0 \quad (7)$$

is of degree equal to the degree of R in t . If τ is an arbitrary value and if the roots of

$$R(t, \tau) = 0$$

are

$$t = \tau, \tau', \tau'', \dots,$$

then for every index h the functions

$$\Phi_h(t, \tau), \quad \Phi_h(t, \tau'), \quad \Phi_h(t, \tau''), \dots$$

have the same common divisor. The equations

$$R(t, \tau) = 0, \quad R(t, \tau') = 0, \quad R(t, \tau'') = 0, \dots$$

therefore have the same roots. Hence the roots of the last displayed equation in ξ are precisely

$$\xi = \tau, a\tau', \tau'', \dots$$

On the other hand, from $\Phi_h(\tau', \tau) = 0$ one obtains

$$\frac{\varphi_h(\tau)}{\psi_h(\tau)} = \frac{\varphi_h(\tau')}{\psi_h(\tau')} = \frac{\varphi_h(\tau'')}{\psi_h(\tau'')} = \dots$$

Therefore each u_h is a symmetric function of the roots of the equation above and is rationally expressible by its coefficients, hence rationally expressible by ϑ . On interchanging again t and τ , we obtain formulae of the form

$$u = f(\vartheta), \quad u_1 = f_1(\vartheta), \quad \dots, \quad u_{\nu-1} = f_{\nu-1}(\vartheta), \quad (8)$$

where $f, f_1, \dots, f_{\nu-1}$ are rational functions. This proves the theorem.

§108. Resolvents with One Parameter

We return to the hypotheses of §106 and denote by

$$u, u_1, \dots, u_{\nu-1}$$

a system of rational functions of $x_0, x_1, \dots, x_{n-1}$ which arise from one of them by the permutations of the alternating group and which are roots of the equation

$$F(u, z) = 0, \tag{9}$$

where z is a rational function of the x 's remaining unchanged under the alternating group.

We prove a second auxiliary theorem.

2. If the rational function

$$\Psi(u, u_1, \dots, u_{\nu-1})$$

vanishes identically after substituting for $x_0, x_1, \dots, x_{n-1}$ certain rational functions of one variable t , and if under this substitution z does not become independent of t , then Ψ also vanishes identically as a function of the independent variables $x_0, x_1, \dots, x_{n-1}$.

The Galois group of the equation $F(u, z) = 0$ over the field of rational functions in z consists of certain permutations of the roots $u, u_1, \dots, u_{\nu-1}$. Perform these permutations in the function Ψ and form the product

$$\prod \Psi(u, u_1, \dots, u_{\nu-1}) = f(z) \tag{10}$$

over all functions so obtained. This gives a rational function $f(z)$ of z . If, after the substitution in the variable t , one factor of this product vanishes identically and z is not independent of t , then $f(z)$ must be identically zero. Consequently one factor of the product must vanish identically as a function of the x 's. But if one factor vanishes identically, then all factors vanish identically, since in every rational equation among the u 's one may apply all substitutions of the Galois group. This proves the assertion.

Together with Luroth's theorem, this gives the following result.

3. Let $u, u_1, \dots, u_{\nu-1}$ be the roots of an equation depending on one parameter z , and let $u, u_1, \dots, u_{\nu-1}, z$ be rational functions of the independent variables $x_0, x_1, \dots, x_{n-1}$.

Then one may put

$$u = f(\vartheta), \quad u_1 = f_1(\vartheta), \quad \dots, \quad u_{\nu-1} = f_{\nu-1}(\vartheta), \quad (11)$$

where $f, f_1, \dots, f_{\nu-1}$ are rational functions of ϑ and

$$\vartheta = \chi(u, u_1, \dots, u_{\nu-1}) = \psi(x_0, x_1, \dots, x_{n-1}) \quad (12)$$

is a rational function of the u 's, equivalently of the x 's.

Indeed, after replacing $x_0, x_1, \dots, x_{n-1}$ by rational functions of one variable t , in such a way that z does not become independent of t , Luroth's theorem first gives the representation just stated. The relations

$$u_h = f_h(\vartheta) = f_h(\chi(u, u_1, \dots, u_{\nu-1}))$$

are then identities in t , and by the preceding theorem are identities in the variables x .

4. If two variables ϑ, ϑ_1 are each rational functions of the other, then they are linear fractional functions of one another.

For suppose

$$\vartheta_1 = \frac{\varphi(\vartheta)}{\psi(\vartheta)}, \quad \vartheta = \frac{\varphi_1(\vartheta_1)}{\psi_1(\vartheta_1)},$$

where numerator and denominator in each fraction are relatively prime integral functions. If the second relation is arranged as a polynomial equation in ϑ , and if the first relation is substituted, the denominator ψ must divide the leading coefficient and hence must be constant or linear. Similarly φ must be constant or linear. Since not both can be constant, the assertion follows.

Apply this to the function ϑ of theorem 3. If the variables x are subjected to any permutation of the alternating group, then the functions $u, u_1, \dots, u_{\nu-1}$ are also permuted. Hence ϑ is changed into another function ϑ_1 which, by the representation above, is rationally expressible by ϑ . Conversely ϑ is rationally expressible by ϑ_1 . Therefore ϑ and ϑ_1 are linear fractional functions of one another.

This proves:

5. If for a general equation of degree n there exists a resolvent with one parameter, then there is a rational function ϑ of the n variables x which, under every permutation

of the alternating group, is changed into a linear fractional function of itself,

$$\vartheta \longmapsto \frac{a\vartheta + b}{c\vartheta + d}, \tag{13}$$

where a, b, c, d are constants, that is, numbers.

§109. Group of the Resolvents with One Parameter

The function ϑ in theorem 5 of the preceding paragraph is itself the root of a resolvent with one parameter,

$$\Phi(\vartheta, z) = 0.$$

Since each root is rationally expressible by every other root, this equation is normal, and its Galois group is isomorphic to the group of the linear fractional substitutions $\chi(\vartheta)$ occurring in theorem 5. In the identity $\Phi(\vartheta, z) = 0$ all permutations of the alternating group $\mathfrak{A}$ of the variables x may be performed; z remains unchanged, while ϑ is carried into every other root of Φ . Hence the group of the linear substitutions $\chi(\vartheta)$, that is, the Galois group of Φ , is isomorphic, in one or more stages, with the alternating permutation group on n letters. We denote this linear group by L . Since it is finite, it must be one of the polyhedral groups considered in the seventh section.

For simplification we use the theorem of §51, according to which the group L can be transformed so that any chosen non-identity substitution of L becomes a multiplication. Such a transformation of L is simply the replacement of ϑ by a linear fractional function of ϑ , which has the same property as the original function.

Let ϑ_π denote the function obtained from ϑ by applying the permutation π to the variables $x_0, x_1, \dots, x_{n-1}$. We may therefore choose ϑ in such a way that for a fixed but arbitrary permutation π one has

$$\vartheta_\pi = \varepsilon \vartheta.$$

If π is repeated, and if p is its order, then ε is a p -th root of unity.

First let $n = 3$. The alternating group consists of the powers of the cyclic permutation

$$\gamma = (0, 1, 2).$$

We may assume that the corresponding substitution is multiplicative, say $\vartheta_\gamma = \varepsilon \vartheta$, where ε is a third root of unity. Thus ϑ^3 is an alternating, or symmetric, function. As a resolvent with one parameter we obtain the pure equation

$$\vartheta^3 = z.$$

For example, one may take for ϑ the Lagrange resolvent

$$x_0 + \varepsilon x_1 + \varepsilon^2 x_2,$$

which gives the familiar reduction of the cubic equation to a pure equation.

Next consider $n = 4$. The alternating group $\mathfrak{A}$ contains the four-group

$$1, \quad \alpha_1 = (0, 1)(2, 3), \quad \alpha_2 = (0, 2)(1, 3), \quad \alpha_3 = (0, 3)(1, 2),$$

and also eight cyclic permutations of three letters. If

$$\gamma = (0, 1, 2), \quad \alpha_1 = (0, 1)(2, 3),$$

then γ and α_1 generate $\mathfrak{A}$. Put

$$\gamma\alpha_1\gamma = / = (0, 3, 1),$$

so that γ and the indicated generator may also be regarded as generators. Moreover

$$\gamma^2 / = \alpha_1. \tag{14}$$

Choose ϑ so that γ acts by multiplication:

$$\vartheta_\gamma = \varepsilon\vartheta,$$

where ε is a third root of unity. Write

$$\vartheta = \frac{\varphi(x_0, x_1, x_2, x_3)}{\psi(x_0, x_1, x_2, x_3)} \tag{15}$$

with φ, ψ integral forms without common factor. From the identity $\vartheta_\gamma = \varepsilon\vartheta$ it follows that

$$\varphi_\gamma = \varepsilon_1\varphi, \quad \psi_\gamma = \varepsilon_2\psi,$$

where $\varepsilon_1, \varepsilon_2$ are again third roots of unity. Applying any of the α 's gives a second linear fractional expression in ϑ . By comparing numerator and denominator and by eliminating, one obtains a relation of the form

$$h\varphi + h_1\varphi_1 + h_2\varphi_2 = 0.$$

Repeating this relation under the substitutions α_1, α_2 and adding suitable multiples, one concludes that one of the relations

$$\varphi = \pm\varphi_1, \quad \varphi = \pm\varphi_2, \quad \varphi = \pm\varphi_3$$

must hold. Since we may equally well use γ, α_1 , or γ, α_2 , or γ, α_3 as generators, it follows:

1. The function φ is changed by every permutation of the alternating group only by multiplication by a constant factor.

The same proof applies to ψ , and hence also to their quotient ϑ . Thus for every permutation π of the alternating group

$$\vartheta_\pi = \varepsilon \vartheta. \quad (16)$$

If π is a cyclic permutation of three letters, ε is a third root of unity. Since every permutation of $\mathfrak{A}$ is compounded from such cycles, ε is always a third root of unity. If π belongs to the four-group, ε is at the same time a second root of unity and must therefore be 1. Hence:

2. The function ϑ is unchanged by the permutations of the four-group.

The third power ϑ^3 is invariant under the alternating group, and ϑ is therefore the root of a pure cubic equation

$$\vartheta^3 = z.$$

This is the resolvent $\Phi(\vartheta, z) = 0$ in this case; it is a partial resolvent.

The purpose of these considerations is the following theorem.

3. A general equation of degree higher than the fourth has no resolvents with one parameter.

For if $n > 4$, we may regard ϑ as a function of any four of the variables x and apply the preceding result. It follows that ϑ remains unchanged when any pair of transpositions is performed among the variables. But the alternating group is generated by such pairs of transpositions; hence ϑ is invariant under the whole alternating group, that is, it is alternating or symmetric. The equation $\Phi(\vartheta, z) = 0$ then reduces to the identity $\vartheta = z$ and is not a resolvent. If the symmetric group is used instead of the alternating group, the theorem is true a fortiori.

This theorem was first stated by Kronecker, who did not publish a proof. The first proof was given by F. Klein and was then simplified by Gordan; the argument above follows Gordan in its main lines.

§110. The Icosahedral Equation

The preceding result is at first negative for the general equation of degree greater than four: it is not possible, by forming rational resolvents, to reduce the general equation of the fifth degree to an equation with one parameter. If nevertheless one does not give up this aim, there is only one remaining course: the variables

$$x_0, x_1, \dots, x_{n-1}$$

must no longer be regarded as entirely independent, but certain equations among them must be allowed. We then no longer have the general equation of the degree in question, but a more special one, in which the coefficients are not independent. A second question arises: how, and by what irrational functions, can the general equation be reduced to this special one? If this can be done by equations of lower degree, for example by square roots, then a real reduction of the problem has nevertheless been achieved.

We now suppose that the variables $x_0, x_1, \dots, x_{n-1}$ are connected by an arbitrary number of relations

$$A(x_0, x_1, \dots, x_{n-1}) = 0, \quad B(x_0, x_1, \dots, x_{n-1}) = 0, \quad \dots, \quad (17)$$

where $A, B, \dots$ are rational functions of the x 's. These relations are assumed to be so constituted, in number and mutual dependence, that the values of the x 's are not fixed numerically but still vary. Moreover, let a permutation group $\mathfrak{R}$ of the x 's be given. We regard as rational precisely those functions of the x 's which admit the substitutions of $\mathfrak{R}$, so that $\mathfrak{R}$ is the Galois group of the equation whose roots are the $x_0, x_1, \dots, x_{n-1}$. The relations must then be such that they too admit the substitutions of $\mathfrak{R}$.

Under these hypotheses we ask whether two rational functions u, z of the x 's can be found such that an equation

$$F(u, z) = 0 \quad (18)$$

is satisfied for all values of the x 's satisfying the relations, and in which z remains invariant, with respect to the relations, under the substitutions of $\mathfrak{R}$. The function u is to be changed thereby into ν different functions

$$u, u_1, \dots, u_{\nu-1},$$

which are the roots of the equation; if $\nu > 1$, this equation is a resolvent of the equation for the x 's.

In this generality we have no point of attack. We add one further restricting hypothesis: it must be possible to put

$$x_h = \varphi_h(t) \quad (19)$$

with rational functions of a variable t , so that all relations are identically satisfied, and so that at the same time z does not become independent of t .

As an example, which will occupy us later in detail, take five variables x_0, x_1, x_2, x_3, x_4 satisfying

$$\sum x = 0, \quad \sum x^2 = 0. \quad (20)$$

They are then the roots of a principal equation of the fifth degree. The parametrization given in Volume I, §74, shows that our hypothesis is satisfied in this case. Geometrically, if the x and the u are regarded as coordinates of points X and U in spaces of n and ν dimensions, the relations determine for X a manifold of lower dimension. To every X corresponds a U , and by (2) the totality of the U forms a one-dimensional manifold, a curve. The hypothesis says that this curve is rational, or unicursal, or in the terminology of function theory, of genus zero.

Under this hypothesis the preceding arguments remain valid. Thus, if an equation whose roots are the x 's has a resolvent

$$F(u, z) = 0 \quad (21)$$

with one parameter z invariant under $\mathfrak{R}$, then there must be a function ϑ of the variables x which, under every substitution π of $\mathfrak{R}$ and with respect to the relations, undergoes a linear fractional substitution

$$\vartheta_\pi = \frac{a\vartheta + b}{c\vartheta + d}. \quad (22)$$

The roots of the resolvent are rationally expressible by ϑ , and ϑ itself is the root of an equation with one parameter

$$\Phi(\vartheta, z) = 0. \quad (23)$$

If the resolvent is a total resolvent, which always happens when the group is simple, the x 's themselves are rationally expressible by ϑ and the functions belonging to the group $\mathfrak{R}$.

Up to this point there is complete agreement with the results of §108. The restriction encountered in §109 need not occur now, since from the equality of two fractional functions one can no longer infer, as in the case of completely independent variables, the equality of numerator and denominator.

The linear substitutions in ϑ form a group P isomorphic, in one or more stages, with $\mathfrak{A}$. This group must be one of the polyhedral groups. We shall treat in detail only the most interesting case, namely that in which P is the icosahedral group, and we may take it without loss of generality in the normal form of §58.

For this group the three fundamental forms were found in §60. Denote their variables by y_1, y_2 . They are the homogeneous forms of degrees 12, 20, 30:

$$\begin{aligned} f(y) &= y_1 y_2 (y_1^{10} + 11y_1^5 y_2^5 - y_2^{10}), \\ H(y) &= -(y_1^{20} + y_2^{20}) + 228(y_1^{15} y_2^5 - y_2^{15} y_1^5) - 494y_1^{10} y_2^{10}, \\ T(y) &= y_1^{30} + y_2^{30} + 522(y_1^{25} y_2^5 - y_1^5 y_2^{25}) - 10005(y_1^{20} y_2^{10} + y_2^{20} y_1^{10}), \end{aligned} \quad (24)$$

with the relation

$$T^2 + H^3 = 1728f^5. \quad (25)$$

If in the quotient $T^2 : f^5$ we put for $y_1 : y_2$ one of the sixty values ϑ_α obtained from ϑ by icosahedral substitutions, then this quotient has always the same value. It is therefore rationally expressible by the coefficients of the equation for ϑ , hence rationally by z . We set it equal to z and obtain the resolvent

$$T^2 - zf^5 = 0, \quad (26)$$

or, equivalently,

$$H^3 - (1728 - z)f^5 = 0. \quad (27)$$

This equation is homogeneous in y_1, y_2 and of degree 60. If $y_2 = 1$, its sixty roots $y_1 = \vartheta_\alpha$ form the icosahedral equation. Written explicitly in terms of ϑ it is

$$\left(\vartheta^{30} + 522\vartheta^{25} - 10005\vartheta^{20} - 10005\vartheta^{10} - 522\vartheta^5 + 1 \right)^2 = z\vartheta^5(\vartheta^{10} + 11\vartheta^5 - 1)^5. \quad (28)$$

The problem, in the form now reached, is therefore reduced to the question:

Which equations can be solved by the icosahedral equation?

§111. The Resolvents of the Icosahedral Equation

The formulation at the end of the preceding paragraph leads us to the various resolvents of the icosahedral equation. To each divisor of the icosahedral group there corresponds such a resolvent; its degree is the index of the divisor and hence always a divisor of 60. The icosahedral group contains:

1. tetrahedral groups, resolvents of degree 5,
2. dihedral groups D_5 , resolvents of degree 6,
3. dihedral groups D_3 , resolvents of degree 10,
4. cyclic groups C_5 , resolvents of degree 12,
5. four-groups D_2 , resolvents of degree 15,
6. cyclic groups C_3 , resolvents of degree 20,
7. cyclic groups C_2 , resolvents of degree 30.

Since the icosahedral group is simple, all these equations are total resolvents of one another. We restrict ourselves to the most important among them.

The resolvents of lowest degree, the quintic resolvents, belong to the tetrahedral group. To find them we must seek a function of ϑ which remains unchanged under the substitutions of a tetrahedral group but is five-valued in the icosahedral group.

Take the icosahedral group in the normal form of §58 and choose the tetrahedral group Q contained in it. It is generated over the four-group by

$$1, \quad \psi, \quad \chi, \quad \psi\chi, \quad (29)$$

where, for a fifth root of unity ε ,

$$\psi(x) = -\frac{1}{x}, \quad \chi(x) = \frac{\omega x + 1}{x - \omega}, \quad \omega = \varepsilon + \varepsilon^{-1}. \quad (30)$$

Putting $\Theta(x) = \varepsilon x$, the whole icosahedral group decomposes as

$$P = Q + Q\Theta + Q\Theta^2 + Q\Theta^3 + Q\Theta^4, \quad (31)$$

and

$$Q_h = \Theta^{-h} Q \Theta^h \quad (32)$$

are the conjugate tetrahedral groups.

In §56 we met two forms of degrees 6 and 8, say t and ω , invariant under the tetrahedral substitutions when the latter are taken homogeneously with determinant one. In the present normal form their first member is obtained from the equation for the double poles of the tetrahedral group:

$$x = \psi(x), \quad x = \chi(x), \quad x = \psi\chi(x),$$

that is,

$$x^2 + 1 = 0, \quad x^2 - 2\omega x - 1 = 0, \quad \omega x^2 + 2x - \omega = 0,$$

and, using $\omega^2 + \omega = 1$,

$$(x^2 + 1)(x^4 + 2x^3 - 6x^2 - 2x + 1) = 0. \quad (33)$$

Thus the required tetrahedral form of degree 6 in homogeneous variables x_1, x_2 is

$$t(x_1, x_2) = x_1^6 + 2x_1^5x_2 - 5x_1^4x_2^2 - 5x_1^2x_2^4 - 2x_1x_2^5 + x_2^6. \quad (34)$$

The second tetrahedral form is its Hessian determinant. If

$$\Omega(x_1, x_2) = \frac{1}{5^2 \cdot 4^2} (t''_{11}t''_{22} - (t''_{12})^2), \quad (35)$$

then a simple calculation gives

$$\begin{aligned} \Omega(x_1, x_2) = & - (x_1^8 + x_2^8) + (x_1^7x_2 - x_1x_2^7) - 7(x_1^6x_2^2 + x_1^2x_2^6) \\ & - 7(x_1^5x_2^3 - x_1^3x_2^5). \end{aligned} \quad (36)$$

The three icosahedral forms are the forms f, H, T displayed in the preceding paragraph.

From these invariant forms of the tetrahedron one obtains functions of $x_1 : x_2$. Setting $\vartheta = x_1 : x_2$, they are functions unchanged under the fractional tetrahedral substitutions and hence are roots of quintic resolvents of the icosahedral equation. The icosahedral equation itself assumes the two forms

$$H^3 = zf^5, \quad T^2 = z_1f^5, \quad z + z_1 = 1728. \quad (37)$$

If $\varphi(x_1, x_2)$ is any absolute tetrahedral invariant form, then under the substitutions $Q\Theta$ it gives the five forms

$$\varphi_\varepsilon = \varphi(\varepsilon^{-2}x_1, \varepsilon^2x_2). \quad (38)$$

Every symmetric function of these five forms is an invariant of the icosahedral group, and therefore, by §61, can be expressed by the fundamental forms f, H, T . This gives a relatively simple calculation of the resolvents.

The simplest functions to use as roots are

$$r = \frac{t^2}{f}, \quad u = \frac{f^2 t}{T}, \quad v = \frac{f\Omega}{H}. \quad (39)$$

The symmetric fundamental functions of the five functions t_ε are icosahedral invariants of degrees 6, 12, 18, 30. Since degrees 6 and 18 do not occur among the fundamental invariants in §61, one obtains an equation of the form

$$t^5 + a f t^3 + b f^2 t + c T = 0, \quad (40)$$

where a, b, c are numerical coefficients. Determining them from Newton's formulae gives, in the notation above, the quintic equations

$$r(r^2 - 10r + 45)^2 = rz, \quad (41)$$

for r , and

$$u^5 - 10u^3 + 45u - z = 0 \quad (42)$$

for u , after the corresponding normalization. By the substitution

$$y = -\frac{u}{3}, \quad z = 27Z,$$

the latter becomes

$$y^5 + 15y - 10y^3 + 3Z = 0, \quad (43)$$

which has already been encountered in the first volume as a normal form of a quintic equation.

§112. The Principal Resolvent of the Fifth Degree

If the two functions u, v are used simultaneously, one obtains a family of resolvents having the form of the principal equation of the fifth degree. They may be brought directly into agreement with any given principal quintic equation.

Since the icosahedron has no invariants of degrees 8, 14, 16, 22, 28, the sums

$$S(\Omega), \quad S(t\Omega), \quad S(\Omega^2), \quad S(t\Omega^2), \quad S(t^2\Omega^2)$$

vanish. Thus, if

$$Y = \alpha\Omega^2 + \beta t\Omega, \tag{44}$$

then the functions $S(Y)$ and $S(Y^2)$ vanish for arbitrary α, β . This yields an identity of the form

$$Y^5 + 5aY^2 + 5bY + c = 0, \tag{45}$$

where a, b, c are homogeneous functions of degrees 3, 4, 5 in α, β , their coefficients being icosahedral invariants.

The last coefficient follows at once from the product formula:

$$c = -\Pi(\Omega) \Pi(\alpha + \beta t). \tag{46}$$

Using the formula from the preceding paragraph for $\Pi(\lambda - t)$, one obtains

$$c = H^2(\alpha^5 - 10f\alpha^3\beta^2 + 45f^2\alpha\beta^4 + T\beta^5). \tag{47}$$

The two coefficients a, b are obtained most simply from Newton's formulae. The calculation gives

$$\begin{aligned} a &= 8f^2\alpha^3 + T\alpha^2\beta + 12f^3\alpha\beta^2 + fT\beta^3, \\ b &= -fH\alpha^4 + 18f^2H\alpha^2\beta^2 + HT\alpha\beta^3 + 27f^3H\beta^4. \end{aligned} \tag{48}$$

To obtain from this the resolvent of the icosahedral equation, introduce the functions u, v of the preceding paragraph. Put

$$Y = \lambda v + \mu u. \tag{49}$$

Then, if

$$z_1 = 1728 - z, \tag{50}$$

the preceding formulae assume the corresponding normalized form

$$Y^5 + 5aY^2 + 5bY + c = 0. \quad (51)$$

This is a principal equation of the fifth degree. It can represent every principal equation if a, b, c may be taken arbitrarily. Thus, for arbitrary prescribed a, b, c , the three quantities λ, μ, z_1 are to be determined. A remarkable property of this system is that these unknowns are rationally expressible by a, b, c and by the discriminant of the quintic.

Weber derives this by eliminating the parameters. From the formulae one obtains first

$$z_1(7b + c) = \lambda^2 a, \quad (52)$$

and another linear relation between a, b, c and the remaining parameter. Squaring, subtracting, and using $z + z_1 = 1728$, one is led to the quadratic equation

$$\lambda^2(a^4 + abc - b^3) - \lambda(11a^2b - ac^2 + 2b^2c) - 27a^2c + 64ab^2 - bc^3 = 0. \quad (53)$$

Its discriminant is

$$\Delta = a^2(108a^3c - 135a^2b^2 + 90abhc^2 - 320ab^2c + 256b^3 + c^4), \quad (54)$$

which is, up to the factor displayed in Weber's normalization, the discriminant D of the principal quintic. Thus

$$\Delta = a^2 D. \quad (55)$$

Consequently λ is obtained with the square root of the discriminant, and after λ is known the ratio of the remaining parameters follows rationally; the last unknown is then obtained rationally as well.

Thus, besides the square root of the discriminant, which is rationally expressible by the roots, a further occurrence of $\sqrt{5}$ appears.

§113. Resolvents of the Sixth Degree

To find the resolvents of degree six of the icosahedral equation, begin with one of the dihedral groups D_5 contained in the icosahedral group. The representation of the icosahedral group in §58 gives the decomposition into cosets directly. If

$$Q = \Theta^r \psi, \quad r = 0, 1, 2, 3, 4, \quad (56)$$

is the dihedral group with which we begin, then the full icosahedral group may be written

$$P = Q + Q\chi + Q\chi\Theta + Q\chi\Theta^2 + Q\chi\Theta^3 + Q\chi\Theta^4. \quad (57)$$

If U_∞ is a function invariant under the substitutions of Q , and if it is changed by

$$\chi, \quad \chi\Theta, \quad \chi\Theta^2, \quad \chi\Theta^3, \quad \chi\Theta^4$$

into

$$U_0, U_1, U_2, U_3, U_4,$$

then

$$U_\infty, U_0, U_1, U_2, U_3, U_4$$

are the roots of a resolvent of the sixth degree.

One may use the invariant forms of the dihedral group and divide by suitable powers of the icosahedral invariants f, H, T . It is enough here to carry through the simplest case. If the substitution χ is applied to the quadratic form in the notation of §58, and the five conjugates are written down, we obtain six forms w ; the first may be written

$$w_\infty = 5x_1^3 x_2^3, \quad (58)$$

and the others in the shape

$$w_r = (\varepsilon^r x_1 + \varepsilon^{-r} x_2)^6 \quad (59)$$

with the appropriate signs and normalizing factors used in §58.

The elementary symmetric functions of these six forms are icosahedral invariants. From comparison of degrees and of one coefficient each Weber obtains the power sums, whence

$$\Pi(w) = 5f^2. \quad (60)$$

Newton's formulae then give the equation

$$w^6 - 10fw^3 + Hw + 5f^2 = 0. \quad (61)$$

If

$$U = \frac{Tw}{fH}, \quad (62)$$

then one obtains the sextic resolvent of the icosahedral equation

$$U^6 - 10zU^3 + z^2U + 5 = z^2. \quad (63)$$

The six roots satisfy the Jacobi relations. If they are denoted by

$$U_\infty, U_0, U_1, U_2, U_3, U_4,$$

then the linear combinations with coefficients $1, \varepsilon, \varepsilon^2, \varepsilon^3, \varepsilon^4$ have the form displayed by Jacobi, and the equations of degree six whose roots satisfy these relations are called Jacobi equations of the sixth degree.

§114. On the Solution of the Icosahedral Equation by Transcendental Functions

We finally indicate, without proof and without detailed development, the relations between the preceding algebraic investigations and the theory of certain transcendent functions, especially elliptic functions and hypergeometric series. From these relations the solution of the icosahedral equation, and therefore of all its resolvents, by transcendent functions follows. This lies outside algebra proper, but stands to it in roughly the same relation as the use of logarithms in the computation of roots, or the trisection of angles in the solution of cubic equations.

Let ω be a variable whose imaginary part is positive, so that

$$q = e^{2\pi i \omega} \tag{64}$$

has absolute value less than one. Weber introduces three functions of ω , denoted in the source by combinations of infinite products and series, from which the modular invariant is formed. In modern notation these are the theta-constant expressions which transform into one another under the substitutions of the modular group. From them one constructs a function $J(\omega)$ and a sextic equation whose Galois group is the same as that of the Jacobi sextic resolvent.

The roots of this sextic may be put in correspondence with the quantities $v_\infty, v_0, \dots, v_4$ occurring in the previous section. By choosing the fifth root of unity suitably, Weber obtains

$$v_\xi = -\eta_\xi(\omega), \quad \xi = \infty, 0, 1, 2, 3, 4, \tag{65}$$

where the η_ξ are explicit theta functions. Thus, once ω has been determined from the transcendent equation giving the prescribed value of z , the formulae of the preceding paragraph give the complete solution of the sextic icosahedral resolvent.

In particular, for $\xi = 0, 1, 2, 3, 4$ Weber writes

$$\eta_\xi(\omega) = \sum (-1)^n e^{\pi i \omega (6n+1)^2} \tag{66}$$

with the congruence classes of n divided according to their residues modulo 5. From the decomposition of this series one recognizes directly that the five quantities satisfy

Jacobi's relations. The root $x = x_1 : x_2$ of the icosahedral equation is then represented as a quotient of two such theta series; one of Weber's forms is

$$x = \frac{\sum (-1)^n e^{\pi i \omega (6n+1)^2}}{\sum (-1)^n e^{\pi i \omega (6n-1)^2}}, \quad (67)$$

with the summation over the appropriate residue classes.

The remaining roots are obtained by applying the icosahedral substitutions to x . They may also be obtained by replacing the variable ω by an equivalent value.

Weber also records the solution by hypergeometric series. For Gauss' series

$$F(\alpha, \beta, \gamma; \lambda) = 1 + \frac{\alpha\beta}{\gamma}\lambda + \frac{\alpha(\alpha+1)\beta(\beta+1)}{1 \cdot 2\gamma(\gamma+1)}\lambda^2 + \dots,$$

put

$$z = 1728\lambda.$$

Then a root of the icosahedral equation is given by a quotient of two hypergeometric series, one of the forms being

$$\vartheta = \frac{F\left(\frac{1}{60}, \frac{19}{60}, \frac{2}{3}; \lambda\right)}{F\left(\frac{31}{60}, \frac{49}{60}, \frac{4}{3}; \lambda\right)} \quad (68)$$

up to the conventional normalizing factor used in the icosahedral equation. These series converge when $|\lambda| < 1$; for other values they must be replaced by the corresponding analytic continuations.

Fourteenth Section. Groups of Linear Ternary Substitutions

§115. A Ternary Linear Substitution Group of Degree 168

The problem of all possible finite groups of linear substitutions in several variables, especially in three and four dimensions, was treated by C. Jordan. As in the binary case, Jordan found a finite number of possible types. We shall not follow these general investigations here; we restrict ourselves to one example of a ternary group.

In the section on congruence groups we met a series of simple groups. The first, of order 60, proved to be isomorphic with the icosahedral group. We now consider the next one, of order 168, which was denoted by L_7 , and we shall construct a ternary linear substitution group isomorphic with it.

We are led to this by the theorem of §72. We seek four elements

$$\tau, \chi, \omega, \Theta$$

which satisfy the composition relations stated there. First take a substitution τ in normal form

$$\tau = \begin{pmatrix} \varepsilon & 0 & 0 \\ 0 & \varepsilon^2 & 0 \\ 0 & 0 & \varepsilon^4 \end{pmatrix}, \quad (69)$$

where ε is a primitive seventh root of unity. The product of the three diagonal factors is one.

We next determine χ from the condition

$$\chi\tau\chi^{-1} = \tau^2. \quad (70)$$

This leads to a cyclic permutation of the three variables. Up to a simultaneous transformation of the whole group, we may put

$$\chi = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad \chi^{-1} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}. \quad (71)$$

The substitutions $\tau^r\chi^s$ then form a group of order 21.

To determine ω , use the relation

$$\omega\chi\omega = \chi^{-1}, \quad (72)$$

or, equivalently, the relation given in §72. If

$$\omega = \begin{pmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{pmatrix}, \quad (73)$$

then the condition that ω have order two gives

$$\begin{aligned} \alpha^2 + \beta^2 + \gamma^2 &= 1, \\ \alpha\beta + \beta\gamma + \gamma\alpha &= 0. \end{aligned} \quad (74)$$

Together with the determinant condition one obtains

$$\alpha + \beta + \gamma = -1, \quad (75)$$

and hence

$$\alpha = \beta\gamma - \alpha^2, \quad \beta = \gamma\alpha - \beta^2, \quad \gamma = \alpha\beta - \gamma^2. \quad (76)$$

The coefficients are expressed by the seventh root ε as

$$\alpha = h(\varepsilon^3 - \varepsilon^{-3}), \quad \beta = h(\varepsilon^2 - \varepsilon^{-2}), \quad \gamma = h(\varepsilon - \varepsilon^{-1}), \quad (77)$$

where

$$h = \frac{1}{\sqrt{-7}} \quad (78)$$

with the sign chosen according to the chosen primitive seventh root. Equivalently,

$$\alpha = -\frac{2\sin(6\pi/7)}{\sqrt{7}}, \quad \beta = -\frac{2\sin(4\pi/7)}{\sqrt{7}}, \quad \gamma = -\frac{2\sin(2\pi/7)}{\sqrt{7}}. \quad (79)$$

The useful relations

$$\alpha\varepsilon + \beta\varepsilon^{-2} + \gamma\varepsilon^2 = 1, \quad \alpha\varepsilon^{-1} + \beta\varepsilon^{-4} + \gamma\varepsilon^{-2} = 1 \quad (80)$$

follow from these formulae.

The remaining generator Θ is obtained from the relations in §72. Weber gives it in the form

$$\Theta = \omega\chi\omega\chi^2 \quad (81)$$

after the substitutions are multiplied out. It has order four, and with τ, χ, ω it generates a group of degree 168. The defining relations are exactly those of the congruence group L_7 ; consequently the ternary group so obtained is isomorphic with that simple group.

For the following paragraphs it is not the abstract presentation but the explicit ternary substitutions which will be used: the diagonal element τ , the cyclic permutation χ , the symmetric involution ω with the coefficients α, β, γ , and the fourth-order element Θ generated from them.

§116. Poles and Axes of Ternary Groups

A ternary linear substitution

$$y_i = \sum_j a_{ij} x_j \quad (82)$$

may be regarded as a projective collineation of the plane. To a point with homogeneous coordinates (x_1, x_2, x_3) it assigns a point (y_1, y_2, y_3) ; to every line it assigns a line. If two points are joined by a line, then the images of the two points are joined by the image of that line.

If a second substitution S is used to transform A into

$$A' = S^{-1}AS,$$

this has the same effect as changing the system of coordinates. Composition is preserved by such transformations, and every group is thereby transformed into an isomorphic group.

The most important points for a substitution A are those left fixed by it. If

$$A = \begin{pmatrix} \alpha & \beta & \gamma \\ \alpha' & \beta' & \gamma' \\ \alpha'' & \beta'' & \gamma'' \end{pmatrix}, \quad (83)$$

then a fixed point is determined by the equations

$$\begin{aligned} \lambda x_1 &= \alpha x_1 + \beta x_2 + \gamma x_3, \\ \lambda x_2 &= \alpha' x_1 + \beta' x_2 + \gamma' x_3, \\ \lambda x_3 &= \alpha'' x_1 + \beta'' x_2 + \gamma'' x_3. \end{aligned} \quad (84)$$

The possible values of λ are the roots of the characteristic equation. For each root one obtains, unless exceptional coincidences occur, a fixed point. Such fixed points Weber calls the *poles* of the substitution.

Correspondingly, lines may be left invariant. In the ordinary case of three distinct poles, the sides of the triangle formed by the poles are invariant lines. Weber calls these invariant lines the *axes* of the substitution.

Special cases occur. Two or all three poles may coincide. More important is the case in which infinitely many points are fixed. If the substitution is not merely a scalar

multiplication, these fixed points lie on a line. Such a line is called a *principal axis*. A substitution with a principal axis has, besides it, one pole, which in exceptional cases may lie on the principal axis; in a finite group this exceptional case does not occur except for scalar multiplication.

If the principal axis is the line $x_1 = 0$, then the substitution has the form

$$y_1 = \alpha x_1, \quad y_2 = \beta x_2, \quad y_3 = \beta x_3$$

after a suitable choice of coordinates, with $\alpha \neq \beta$. A line is then invariant either if it is the principal axis itself or if it passes through the remaining pole. Thus:

If a substitution has a pole and a principal axis, then, besides the principal axis, the invariant lines are exactly the lines passing through the pole.

Collecting the possibilities, ternary linear substitutions may have:

1. three poles,
2. two poles,
3. one pole,
4. one principal axis and one pole,
5. one principal axis only,

with scalar multiplication as the limiting case in which every point is fixed. The type is invariant under transformation.

If a point is fixed by a substitution A , then it is fixed by all powers of A . Conversely, a power of A may have a principal axis even when A has three poles: then two poles of A lie on the principal axis of the power, and the third pole is the isolated pole of that power. If A has finite order μ , its eigenvalues are μ -th roots of unity; the first power of A which has a principal axis corresponds to the first coincidence among suitable powers of two eigenvalues.

§117. Application to the Group G_{168} . Sevenfold Poles

We apply the preceding generalities to the ternary group constructed in §115. It is convenient first to arrange the elements of the abstract group according to their periods. The elements of order seven, together with the identity, are divided into eight cyclic groups of seven elements each. The elements of order three are divided into twenty-eight cyclic groups of three elements each. The elements of orders four and two are divided into twenty-one cyclic groups of four elements; each of these contains one element of order two and two elements of order four.

Thus the group contains

48 elements of order 7, 56 elements of order 3, 42 elements of order 4, 21 elements of order 2.

If a point of the projective plane is fixed by v substitutions of the group, the identity being included, we shall call it a v -fold pole. This is the same terminology as in the discussion of binary groups, but now the poles are points in the projective plane.

For the determination of all poles and axes it is enough to examine one representative of each cyclic class, since all the others arise by transformation with elements of the group. We begin with the substitution of seventh order, denoted in §115 by r . Since, in the chosen coordinates, r is diagonal, the vertices of the coordinate triangle are its three distinct poles. All non-trivial powers of r have the same three poles.

The cyclic permutation χ carries these three vertices cyclically into one another. On the other hand, the substitutions denoted by $\omega, \omega^2, \Theta, \omega\Theta, \omega^2\Theta, \omega\Theta^2, \omega^2\Theta^2$, when applied to the coordinate triangle, produce triangles which are wholly different from it. For example, under one of these transformations the triangle with vertices

$$(1 : 0 : 0), \quad (0 : 1 : 0), \quad (0 : 0 : 1)$$

passes into a triangle whose vertices have coordinates proportional to

$$(\alpha : \beta : \gamma), \quad (\beta : \gamma : \alpha), \quad (\gamma : \alpha : \beta).$$

It follows that no one of these poles can have a stabilizer larger than the cyclic group of order seven which has already been found.

There are therefore eight triples of such sevenfold poles, corresponding to the eight cyclic subgroups of order seven. These twenty-four points are all distinct. Moreover

any one of them can be transformed into any other by a substitution of the full group G_{168} . We have therefore the theorem:

There are twenty-four sevenfold poles. They divide into eight triples, corresponding to the eight cyclic groups of order seven, and the full group G_{168} is transitive on the twenty-four points.

Each triple may be regarded as a triangle. The coordinate triangle is only the first of the eight; the remaining seven are its transforms. Later these same triangles will appear as triangles of inflection points of the fundamental quartic. The passage from the algebraic substitution r to its three projective poles is therefore not only a calculation of fixed points; it is the first step in the construction of the whole geometric configuration belonging to G_{168} .

The dual lines of these triangles are obtained from the contragredient group. Whenever a substitution has three distinct poles, it also has three axes, namely the lines joining the poles pairwise in the dual description. The systems of poles and axes will be treated together in the following paragraphs.

§118. The Principal Axes

We pass next to the substitutions of fourth and second order. As representatives we choose Θ and Θ^2 . The poles of a ternary substitution are found by solving the characteristic equation and substituting the eigenvalues in the equations

$$Ax = \lambda x.$$

For Θ , after inserting the formulae of §115 and using the relations among α, β, γ , the characteristic equation reduces to

$$\lambda^3 - \lambda^2 + \lambda - 1 = 0.$$

Its roots are

$$\lambda = 1, \quad \lambda = i, \quad \lambda = -i.$$

Thus the substitution Θ has three distinct poles.

For Θ^2 the corresponding cubic has the form

$$(\lambda + 1)^2(\lambda - 1) = 0.$$

Here the double root $\lambda = -1$ gives not a single pole but a whole line of fixed points: this line is a principal axis. The simple root $\lambda = 1$ gives an isolated pole not lying on that line.

Putting $\lambda = -1$ into the equations for Θ^2 , Weber obtains a line which, in his notation, is of the form

$$A = \alpha\gamma x_1 + \beta\gamma\epsilon^m x_2 + \alpha\beta\epsilon^n x_3 = 0,$$

with definite exponents depending on the representative chosen. Applying the group to this line gives all principal axes. Since the group has order 168, and the stabilizer of a principal axis has order 8, there are exactly

$$\frac{168}{8} = 21$$

principal axes.

Let A be one of these axes. Weber determines its stabilizer directly. The line is fixed by Θ^2 , because it is the principal axis belonging to the double root; it is also fixed by

a second substitution, denoted ω , for substitution in the equation of A changes its left-hand side only by a factor -1 . Consequently the whole stabilizer of A is a group of order eight, which Weber denotes by G_A . In symbolic form it is generated by the substitutions corresponding to ω and to powers of Θ .

Among these eight substitutions only two, the identity and Θ^2 , leave every point of A fixed. The others have only two fixed points on A . Thus the same line is the principal axis for exactly one involution in the stabilizer, but it also contains poles belonging to substitutions of fourth order.

The isolated pole of Θ^2 , obtained from the simple root $\lambda = 1$, has coordinates proportional to

$$x_1 : x_2 : x_3 = \alpha\gamma\epsilon^{-3} : \beta\gamma\epsilon : \alpha\beta\epsilon^2.$$

Call this point temporarily a . It is fixed not only by Θ^2 but by the whole group G_A of order eight; conversely its orbit under G_{168} has twenty-one elements. Hence a is exactly an eightfold pole. We therefore obtain twenty-one eightfold poles.

The four substitutions of fourth order in G_A have their poles on the principal axis A . Their isolated poles, transformed from a , are again eightfold poles. On each principal axis lie four eightfold poles, and through each eightfold pole pass four principal axes. This follows equally from the stabilizer calculation and from the incidence count

$$21 \cdot 4 = 21 \cdot 4.$$

The two remaining fixed points of Θ on the principal axis are not eightfold. They are poles of substitutions of order four and give the system of fourfold poles. There are forty-two of these points; each principal axis contains two of them. Thus the principal axes already account for the systems

$$21P_8, \quad 42P_4.$$

The notation P_v indicates a v -fold pole.

The result of this paragraph is the following concise statement. The group has twenty-one principal axes. Each principal axis contains four eightfold poles and two fourfold poles. Through each eightfold pole pass four principal axes. The principal axes and the eightfold poles are thus paired in a symmetric incidence system, while the fourfold poles appear in pairs on each axis.

§119. The Threefold and Sixfold Poles

It remains to examine the substitutions of third order and the related sixfold poles. A representative is the cyclic substitution χ , or a conjugate of it. Its characteristic equation has the cube roots of unity as roots. Accordingly the corresponding eigenvectors give three fixed points in the projective plane.

Two of these points form a pair belonging to the substitution of third order. Their coordinates may be written, in a suitable normalization, in the form

$$x_1 = \rho x_2 = \rho^2 x_3,$$

or in the two conjugate forms obtained by replacing ρ by ρ^2 , where ρ is a primitive cube root of unity. These points are fixed by the cyclic group generated by the substitution, but by no larger subgroup. Hence they are threefold poles.

Weber proves this by excluding the other possibilities. If a larger group fixed one of these two points, the additional substitutions would have to be of orders seven, four, or two. But the coordinates of the sevenfold and fourfold poles are rational functions of the seventh root ϵ , while the coordinates here contain the independent cubic irrationality ρ . A substitution of second order would put the point on a principal axis, and the explicit equations of the principal axes show that this cannot occur. Therefore the stabilizer has exactly three elements.

Thus there are

$$\frac{168}{3} = 56$$

threefold poles. They are paired, two by two, according to the cyclic subgroup of order three to which they belong. Equivalently, the twenty-eight cyclic subgroups of order three give twenty-eight pairs of threefold poles. The full group is transitive on the fifty-six points.

Let

$$T = x_1 + x_2 + x_3.$$

The equation $T = 0$ is the line joining one such pair of threefold poles. The line remains fixed under the substitution χ and changes sign under a suitable involution. Under the full group it gives twenty-eight distinct lines, which Weber denotes collectively by the lines T .

On the line $T = 0$ lie three of the eightfold poles described in the preceding paragraph. They are obtained from the representative eightfold pole by the substitutions belonging to the same normalizer. Thus:

Each line T contains three eightfold poles and two threefold poles.

Since the group acts transitively on the T -lines, this statement holds for all twenty-eight of them.

The third pole of the substitution χ , call it t , is of a different kind. It is fixed by six substitutions, namely by the identity, by the two powers of χ , and by three further substitutions which, together with these, form a group of order six. To show that the stabilizer is no larger, Weber again uses the list of possible orders and the already determined axes. The point t lies on three principal axes, those corresponding to the three involutions in the stabilizer. If another involution fixed t , it would have to add a fourth principal axis through t , but the explicit equations show that no such fourth axis passes through it.

Therefore the point t is exactly sixfold. There are

$$\frac{168}{6} = 28$$

sixfold poles. They form a transitive system. Through each sixfold pole pass exactly three principal axes. Conversely, since there are twenty-one principal axes, each containing the same number of sixfold poles, the incidence count gives

$$28 \cdot 3 = 21 \cdot 4,$$

so that each principal axis contains four sixfold poles.

The twenty-eight lines T and the twenty-eight sixfold poles t are thus linked to the same order-three subgroups, but the sixfold pole does not lie on the corresponding line T . This will be important in the configuration below.

§120. The Configuration of the Group G_{168}

We now collect the results in a single configuration. No new calculation is involved; the purpose is only to make the statements of the last paragraphs more transparent.

The configuration consists first of

21 lines A (the principal axes),

21 points a (the eightfold poles),

28 lines T , 28 points t (the sixfold poles).

The whole system is transformed into itself by the 168 substitutions of the group.

The principal incidence statements are:

on each line A lie four points a , through each point a pass four lines A ,

on each line T lie three points a , through each point t pass three lines A .

Moreover the points a and t together form the complete system of intersections of the twenty-one lines A . Indeed, twenty-one lines have

$$\frac{21 \cdot 20}{2} = 210$$

pairwise intersections when multiplicities are counted. At a point a four lines meet; this contributes $\binom{4}{2} = 6$ pairwise intersections, and at a point t three lines meet; this contributes $\binom{3}{2} = 3$. Since

$$21 \cdot 6 + 28 \cdot 3 = 126 + 84 = 210,$$

all intersections are accounted for.

From the same count one obtains the remaining incidences. Since through every point t pass three lines A , and since all lines A are conjugate, the number of points t lying on a fixed line A is

$$\frac{28 \cdot 3}{21} = 4.$$

Similarly every point a lies on four of the lines T . Finally no point t lies on a line T ; otherwise the incidence counts and the stabilizers would be incompatible.

If the v -fold poles are denoted by P_v , the systems of poles are as follows:

system	number of points
P_8	21
P_6	28
P_4	42
P_3	56
P_7	24
P_2	infinitely many

The twofold poles are all points on the principal axes, except for the special points already counted as P_4, P_6, P_8 . On each principal axis lie two P_4 's; on each line T lie two P_3 's. The sevenfold poles do not lie on the principal axes, and it will follow from the curve theory that they do not lie on the lines T either.

The corresponding system of axes is analogous. Every line through an eightfold pole is an axis; among them are the principal axes, four of the lines T , and two further lines joining that eightfold pole to suitable fourfold poles. It is natural, therefore, to regard the eightfold poles as the principal poles. There are also twenty-eight pairs of connecting lines from a sixfold pole to the two associated threefold poles; and there are twenty-four lines joining paired sevenfold poles, represented by the sides of the coordinate triangle.

This configuration is the projective image of the abstract structure of G_{168} . The incidence relations determine the later invariant curves almost without further calculation: if an invariant curve contains one point of any of these systems, it contains the whole system; if a line is tangent or bitangent in one place, all its conjugates have the corresponding contact.

§121. Invariant Curves of the Group G_{168}

Let $\varphi(x_1, x_2, x_3)$ be a homogeneous ternary form of degree m . Under the substitutions of G_{168} it generally passes into 168 different forms. For special forms this number may be smaller. In that case the substitutions by which φ is unchanged form a subgroup G' of G_{168} ; the number of distinct forms obtained from φ is then the index of G' . The form φ is an absolute invariant of G' , or at least a relative invariant if it is multiplied by a constant factor.

The equation

$$\varphi = 0$$

represents a curve in the projective plane whose coordinates are x_1, x_2, x_3 . The substitutions of the group map this curve to its conjugate curves. If not all the images are distinct, then the curve is left invariant by a subgroup. In particular, if φ is invariant under the whole group, the curve is mapped onto itself by all 168 collineations.

Projective properties are preserved by these collineations. If a line is a tangent, an inflection tangent, or a bitangent of a curve, then every image of the line has the same relation to the corresponding image of the curve. Likewise double points, inflection points, cusps, points of contact of bitangents, and all similar projective properties pass to their images.

We are chiefly concerned with forms which remain unchanged under the whole group. Since the group is simple, the invariants that occur here are absolute. If $\Phi(x_1, x_2, x_3)$ is such an invariant, the curve

$$\Phi = 0$$

will be called an invariant curve of the group. A point on an invariant curve carries with it all its images. In general a point has 168 distinct images. The number is smaller only when the point is one of the poles; then the possible orbit sizes are

$$24 \text{ for } P_7, \quad 21 \text{ for } P_8, \quad 28 \text{ for } P_6, \quad 42 \text{ for } P_4, \quad 56 \text{ for } P_3,$$

and, for a connected system of twofold poles, 84. Thus, if one pole of a given system lies on an invariant curve, all poles of that system lie on it.

The subsequent determination of the invariants is therefore guided by the configuration. The degree of a curve and Bezout's theorem control which systems of poles it can contain. Conversely, an invariant curve of small degree is often forced once its intersections with the fundamental configuration are known.

§122. The First Invariant of the Group G_{168} and the Fundamental Curve

We now form the invariants explicitly. It is enough to require invariance under the three generators r, χ, ω introduced in §115, since the whole group is generated by them. The substitution χ cyclically permutes the variables, while r is diagonal:

$$x_1, x_2, x_3 \longmapsto \epsilon x_1, \epsilon^2 x_2, \epsilon^4 x_3.$$

If a monomial

$$x_1^{h_1} x_2^{h_2} x_3^{h_3}$$

occurs in an invariant, invariance under r requires

$$h_1 + 2h_2 + 4h_3 \equiv 0 \pmod{7}.$$

Invariance under χ requires that, together with this monomial, its two cyclic transforms occur with corresponding coefficients, unless $h_1 = h_2 = h_3$.

No invariant of degree smaller than three can satisfy the congruence. In degree three the only possibility is $x_1 x_2 x_3$, but this product is not left invariant by ω . Thus the least possible degree is four. Solving

$$h_1 + h_2 + h_3 = 4, \quad h_1 + 2h_2 + 4h_3 \equiv 0 \pmod{7}$$

in non-negative integers gives only the three cyclic possibilities

$$(h_1, h_2, h_3) = (3, 1, 0), \quad (0, 3, 1), \quad (1, 0, 3).$$

Hence the only form of fourth degree invariant under r and χ , apart from a constant factor, is

$$f(x_1, x_2, x_3) = x_1^3 x_2 + x_2^3 x_3 + x_3^3 x_1.$$

It remains to verify invariance under ω . Weber substitutes

$$x_1 = \alpha y_1 + \beta y_2 + \gamma y_3,$$

$$x_2 = \beta y_1 + \gamma y_2 + \alpha y_3,$$

$$x_3 = \gamma y_1 + \alpha y_2 + \beta y_3,$$

where α, β, γ have the values fixed in §115. If

$$F(y_1, y_2, y_3) = f(x_1, x_2, x_3),$$

then the second derivatives of F , computed with the relations among α, β, γ , reduce to exactly the second derivatives of

$$y_1^3 y_2 + y_2^3 y_3 + y_3^3 y_1.$$

By Euler's theorem for homogeneous functions it follows that

$$F(y_1, y_2, y_3) = y_1^3 y_2 + y_2^3 y_3 + y_3^3 y_1.$$

Thus f is indeed an invariant of the group. Apart from multiplication by a constant, it is the unique invariant of degree four.

The curve

$$f = 0$$

is the fundamental curve of the group. It is a plane quartic. The side $x_1 = 0$ of the coordinate triangle cuts the curve in three coincident points at one vertex and in a fourth point at another vertex. Thus the vertices of the coordinate triangle are inflection points, and the sides of the coordinate triangle are inflection tangents. With the usual notation, side 1 is the inflection tangent at vertex 2, side 2 at vertex 3, and side 3 at vertex 1.

Since the vertices of the coordinate triangle are sevenfold poles, all the sevenfold poles lie on the fundamental curve. They are all inflection points, and they form eight inflection triangles, corresponding to the eight cyclic groups of order seven.

The threefold poles also lie on the fundamental curve. If ρ is a primitive cube root of unity and one substitutes

$$x_2 = \rho x_1, \quad x_3 = \rho^2 x_1$$

(or the conjugate relation), the equation $f = 0$ is satisfied because

$$1 + \rho + \rho^2 = 0.$$

Their connecting line

$$T = x_1 + x_2 + x_3 = 0$$

therefore meets the quartic at the two threefold poles. In fact the four intersections coalesce two by two; the line T is a bitangent of the fundamental curve. Consequently all twenty-eight lines T are bitangents, and their points of contact are precisely the fifty-six threefold poles.

The sixfold and eightfold poles do not lie on the fundamental curve. For the sixfold pole one may use coordinates proportional to $(1 : 1 : 1)$, which gives $f = 3$. For the eightfold pole Weber substitutes the coordinates found in §118 and uses the relations among α, β, γ ; the resulting value is non-zero.

It remains to determine the intersections of the principal axes with the quartic. Since all intersections of principal axes with one another are the points P_8 and P_6 , and neither of these systems lies on the curve, no two principal axes meet on the curve. If a fourfold pole on a principal axis lay on the quartic, its conjugate fourfold pole on the same axis would also lie there; then the axis would have to be a bitangent. But the twenty-eight lines T already give all bitangents of a nonsingular quartic. Hence the fourfold poles do not occur there. The intersections of a principal axis with the quartic are therefore twofold poles.

We denote these special twofold poles by P_2 . The distinguished point systems are now:

$$24P_7, \quad 56P_3, \quad 84P_2 \quad \text{on the curve } f = 0,$$

and

$$21P_8, \quad 42P_4, \quad 28P_6 \quad \text{not on the curve.}$$

All other points of the curve are transformed into 168 distinct points.

This also proves that the fundamental curve has no double point and cannot decompose into curves of lower degree. For if it had a double point, all its images would be double points; a quartic cannot have so many double points unless it contains a repeated component. But then f would have to be the square of a quadratic invariant, and no quadratic invariant exists. Directly, the partial derivative equations force

$$x_1 = x_2 = x_3 = 0,$$

which is not a point of the projective plane. Hence the quartic is nonsingular.

For brevity Weber calls a system of points, any one of which can be transformed into any other by substitutions of the group, a system of connected points.

§123. The Higher Invariants

Further invariants are obtained from the quartic by forming covariants. The first is the Hessian covariant. With Weber's normalization it is

$$\Delta = \frac{1}{54} \det \left(\frac{\partial^2 f}{\partial x_i \partial x_j} \right),$$

an invariant of degree six. Its developed expression is again invariant under the cyclic permutations and under the diagonal substitution. The curve $\Delta = 0$ contains the vertices of the coordinate triangle; consequently, by the group action, it contains all sevenfold poles. Since the intersections of $f = 0$ and $\Delta = 0$ have total number

$$4 \cdot 6 = 24,$$

these sevenfold poles form the complete system of intersections of the fundamental quartic with its Hessian curve.

A second covariant, of degree fourteen, is obtained from the determinant described in the general theory of covariants. Weber denotes it by C . In determinant form it is the bordered determinant built from the second derivatives of f and the first derivatives of Δ . Written symbolically,

$$C = \det \begin{pmatrix} f_{11} & f_{12} & f_{13} & \Delta_1 \\ f_{21} & f_{22} & f_{23} & \Delta_2 \\ f_{31} & f_{32} & f_{33} & \Delta_3 \\ \Delta_1 & \Delta_2 & \Delta_3 & 0 \end{pmatrix},$$

up to the constant factor chosen by Weber. When developed, it is a cyclic-sum expression in the variables x_1, x_2, x_3 ; Weber writes the expansion explicitly and uses the notation Σ for the sum of the three cyclic terms obtained from the first term by cyclic permutation of the variables.

For the subsequent geometry only two features of this expression are immediately needed. First, from the leading term one sees that the curve $C = 0$ does not pass through the vertices of the coordinate triangle. Second, being an invariant curve, if it contains any one point of a connected pole system, it contains the entire system. It will meet the fundamental curve in the system of threefold poles.

A fourth invariant, of degree twenty-one, is obtained by taking the functional determinant of the three forms f, Δ, C :

$$K = \frac{1}{14} \det \begin{pmatrix} f_1 & \Delta_1 & C_1 \\ f_2 & \Delta_2 & C_2 \\ f_3 & \Delta_3 & C_3 \end{pmatrix}.$$

Only the first terms are needed in the present paragraph to see that this curve also does not pass through the vertices of the coordinate triangle. The full expression is lengthy; Weber refers to Gordan's computation of the complete formula.

The four forms

$$f, \quad \Delta, \quad C, \quad K$$

have degrees

$$4, \quad 6, \quad 14, \quad 21.$$

They will now be shown to generate all invariants of the group. Geometrically, $f = 0$ is the fundamental quartic, $\Delta = 0$ is its Hessian, $C = 0$ is the next distinguished invariant curve, and $K = 0$ is the product of the twenty-one principal axes.

The last assertion is already suggested by the degree. The twenty-one principal axes form an invariant curve of degree twenty-one. In §124 Weber proves that there is, up to a constant factor, only one invariant of degree twenty-one. Hence the invariant K must split into the product of the twenty-one linear forms defining the principal axes.

§124. The Complete System of Invariants

We now prove that the four forms

$$f, \quad \Delta, \quad C, \quad K$$

constitute a complete system of invariants for the group. This means that every invariant of G_{168} can be represented as an integral rational function of these four forms. The proof uses Bezout's theorem: two curves of degrees m and n which have more than mn points in common must have a common component; points of contact are counted twice or with the corresponding higher multiplicity.

Let Φ be an invariant of degree m which does not contain f as a factor. Consider the intersections of the curve $\Phi = 0$ with the fundamental quartic $f = 0$. Among them let the systems P_7, P_3, P_2 occur with multiplicities h_1, h_2, h_3 , and let there also be h_4 arbitrary connected systems of 168 points. Since the intersections with the quartic are counted with multiplicity, one obtains

$$4m = 24h_1 + 56h_2 + 84h_3 + 168h_4,$$

and therefore

$$m = 6h_1 + 14h_2 + 21h_3 + 42h_4. \quad (1)$$

The integers h_1, h_2, h_3, h_4 are non-negative and cannot all vanish.

The smallest value of m is therefore 6. Every invariant of degree six must pass through the twenty-four points P_7 , just as Δ does. If Δ' is another invariant of degree six, choose a constant a so that $\Delta' - a\Delta$ passes through one further point of the quartic. Then, by Bezout's theorem, $\Delta' - a\Delta$ must be divisible by f . The quotient would be an invariant of degree two, which does not exist. Hence Δ' is a constant multiple of Δ .

The same reasoning gives the invariants of degrees twelve and eighteen. If an invariant of degree twelve has the same intersection multiplicities with the fundamental quartic as Δ^2 , then after subtracting a suitable multiple of Δ^2 it must be divisible by f ; the quotient is an invariant of degree eight and is therefore a multiple of f^2 . Thus every invariant of degree twelve has the form

$$a\Delta^2 + bf^3.$$

Similarly every invariant of degree eighteen has the form

$$a\Delta^3 + bf^3\Delta.$$

Consequently all invariants of degrees 6, 12, 18 are functions of f and Δ .

The next possible degree is 14, corresponding to $h_2 = 1$. An invariant of degree fourteen must pass through the fifty-six points P_3 , and these are precisely the complete intersection of the curve $C = 0$ with the fundamental quartic. If C' is another invariant of degree fourteen, choose a so that $C' - aC$ is divisible by f . The quotient is an invariant of degree ten. Since there is no invariant of degree ten except the product $f\Delta$, it follows that every invariant of degree fourteen has the form

$$aC + bf\Delta.$$

Conversely every expression of this form is plainly an invariant of degree fourteen.

For degree twenty, equation (1) allows the system consisting of P_7 and P_3 . The curve must therefore pass through the eighty intersections of $f = 0$ with $\Delta C = 0$. By the same Bezout argument it follows that an invariant of degree twenty is of the form

$$f(a\Delta^2 + bf^3) + c\Delta C,$$

with arbitrary constants a, b, c . Thus there is no new independent invariant of degree twenty.

Consider now invariants of degree twenty-one. They must pass through the eighty-four twofold poles P_2 on the fundamental quartic. If K_1 and K are two invariants of degree twenty-one, a constant a can be chosen so that $K_1 - aK$ is divisible by f . The quotient would be an invariant of degree seventeen, which cannot occur by (1). Therefore, up to a constant factor, there is only one invariant of degree twenty-one.

But the product of the twenty-one linear equations of the principal axes is an invariant of degree twenty-one. Hence this product must be proportional to K . Thus:

The invariant K splits into twenty-one linear factors, and these factors, when set equal to zero, are the principal axes of the group.

We next need a relation. For any constant k the expression

$$Q = K^2 - kC^3$$

is an invariant of degree forty-two. The constant may be chosen so that the curve $Q = 0$ passes through a prescribed connected system of points on the fundamental quartic. Hence, given any invariant Φ not divisible by f , one forms products of such curves Q so as to remove all connected systems occurring in the intersection with $f = 0$, and one subtracts a suitable product of powers of Δ, C, K . The resulting invariant still contains all intersections with $f = 0$, and after one additional choice of a constant it has an extra intersection with the quartic. Bezout's theorem then forces it to be divisible by f . This lowers the degree and permits induction.

This proves:

Every invariant of the group is an integral rational function of the four fundamental invariants f, Δ, C, K .

Since K^2 is of even degree, it must itself be expressible in terms of f, Δ, C . Gordan computed the relation explicitly. In Weber's notation it is

$$\begin{aligned} K^2 = & C^3 - 88f^2\Delta C^2 + 16 \cdot 63f\Delta^4C + 17 \cdot 64f^4\Delta^2C - 256f^7C \\ & + 27 \cdot 64\Delta^7 - 128 \cdot 469f^3\Delta^5 + 43 \cdot 512f^6\Delta^3 - 2048f^9\Delta. \end{aligned}$$

Every term has weighted degree forty-two, when the weights of f, Δ, C are 4, 6, 14.

This relation allows all powers of K except the first to be eliminated from any expression in the invariants. Consequently an invariant of even degree can be expressed rationally through

$$f, \quad \Delta, \quad C,$$

whereas an invariant of odd degree is the product of K and an invariant of even degree. The invariant theory of the ternary group G_{168} is thereby complete.

The next section begins the application of this invariant system to the form problem of the group G_{168} and to the theory of equations of the seventh degree.